Geometric Distribution

Consider the geometric series, $\sum_{x=0}^{\infty} q^x = \frac{1}{1-q}$ if $|q| < 1$

$$\Rightarrow \sum_{x=0}^{\infty} pq^x = 1 \text{ whERe } p = 1 - q.$$

For $0 < q < 1, p + q = 1$, the terms of the geometric series forms a PMF. Hence, the distribution given by the PMF $\{pq^x\}, x = 0, 1, 2, \ldots$ is known as Geometric distribution.

1 Definition

A non-negative integer valued random variable X is said to follow Geometric distribution if its PMF is given by,

$$f(x) = \begin{cases} pq^x & , x = 0, 1, 2, \quad 0 < p < 1 \\ 0, & \text{otherwise} \quad p + q = 1. \end{cases}$$

Remark

Consider a sequence of independent Bernoulli trail with probability of success p. Let X be the no of failure before the 1st success $P(X = r) = P(\underbrace{F \ldots F}_{r} \underbrace{S}_{r+1}) = q^r p$ [Due to independence].

A random variable X that has a geometric distribution is often referred to a discrete waiting time random variable as it represents how long (in terms of failure) one has to wait to get a success.

2 Mean and Variance

$$\sum_{x=0}^{\infty} q^x = \frac{1}{1-q}, \quad 0 < q < 1$$

Differentiating w.r.t q,

$$\sum_{x=0}^{\infty} xq^{x-1} = (-1)(1-q)^{-2}(-1)$$

Again, differentiating w.r.t q,

$$\sum x(x-1)q^{x-2} = (-1)(-2)(1-q)^{-3}(-1)^2, 0 < q < 1$$

$$\Rightarrow \sum x(x-1)q^{x-2} = 2(1-q)^{-3}$$

$$E(X) = \sum xq^x p = qp \sum xq^{x-1} = \frac{qp}{p^2} = \frac{q}{p}$$

$$\begin{aligned} \text{Var}(X) = E(X^2) - E(X)^2 &= E(X(X-1)) + E(X) - E(X)^2 \\ &= \frac{2pq^2}{p^3} + \frac{q}{p} - \frac{q^2}{p^2} \\ &= \frac{2q^2}{p^2} + \frac{q}{p} - \frac{q^2}{p^2} \\ &= \frac{q}{p^2} \end{aligned}$$

$$\therefore E(X) < V(X).$$

Remark

Let Y denote the no of trail required to get the 1st success in a sequence of independent Bernoulli trail.

$$Y = X + 1$$
$$P(Y = y) = P(X + 1 = y) = P(X = y - 1) = pq^{y-1}, y = 1, 2, \ldots$$
$$E(Y) = E(X) + 1 = \frac{q}{p} + 1 = \frac{q+p}{p} = \frac{1}{p}$$
$$V(Y) = V(X) = \frac{q}{p^2}$$

3 Problem

1. In a random sampling WR from a population with N distinct member, let X denote no of drawings needed to get n distinct element the sample. Find $E(X)$ and $V(X)$.

Let $Z_i, i = 1(1)n$ denotes the no of drawing needed to get new distract element when $[i-1]$ distinct element have already obtained.

It is to be noted that $Z_1 = 1$.

$$X = \sum_{i=1}^{n} Z_i = 1 + Z_2 + Z_3 + \cdots + Z_i + \cdots + Z_n$$

Now, Z_i's are independently distributed with pmf,

$P(Z_i = z) = P$ [exactly z drawing are necessary to get new distinct element when $(i-1)$ distinct has already been obtained].

$$= \underbrace{\left(\frac{i-1}{N}\right)\left(\frac{i-1}{N}\right)\cdots\left(\frac{i-1}{N}\right)}_{(z-1)} \frac{N-(i-1)}{N}$$
$$= \left(\frac{i-1}{N}\right)^{z-1}\left(1 - \frac{i-1}{N}\right) = q_i^{z-1} p_i$$

$E(Z_i) = \frac{N}{N-i+1}$
$V(Z_i) = \frac{(i-1)/N}{(N-i+1)^2/N^2} = \frac{N(i-1)}{(N-i+1)^2}$
$E(X) = \sum_{i=1}^{n} E(Z_i) = \sum_{i=1}^{n} \frac{N}{N-i+1}$
$V(X) = \sum_{i=1}^{n} V(Z_i) = \sum_{i=1}^{n} \frac{N(i-1)}{(N-i+1)^2}$

4 Cumulative Distribution Function

$$F_X(x) = P(X \leqslant x) = \sum_{t=0}^{[x]} pq^t$$
$$= p\left[1 + q + q^2 + \cdots + q^{[x]}\right]$$
$$= p\frac{1 - q^{[x]+1}}{1-q} = 1 - q^{[x]+1}$$

Tail function:

$$P(X > x) = 1 - P(X \leqslant x) = 1 - \left(1 - q^{[x]+1}\right) = q^{[x]+1}$$

5 Additional Properties

1. If $X_i \sim \text{Geo}(p), i = 1(1)r, \quad \sum_{i=1}^{r} X_i \sim \text{NB}(r, p)$.
2. $\text{X}_i \sim \text{Geo}(p)$. Find the conditional distribution of $(X_1 \mid X_1 + X_2)$.
   $\frac{P(X_1=x, X_2=y-x)}{P(X_1+X_2=y)} = \frac{P(X_1=x)P(X_2=y-x)}{P(X_1+X_2=y)} = \frac{pq^x pq^{y-x}}{\binom{y+2-1}{y} p^2 q^y} = \frac{1}{(y+1)} \sim DU\{0, 1, 2, \ldots y\}$
3. $X_1, X_2, \ldots X_n \sim \text{Geo}(p)$, find the distribution of $\min(X_i)$, i = 1 (1) n.

$$\begin{aligned} F_Y(y) &= P[Y \leqslant y] = 1 - P(Y > y) \\ &= 1 - P(\min\{Xi\} > y) \\ &= 1 - P(X_1 > y, X_2 > y, \ldots, X_n > y) \\ &= 1 - [P(X > y)]^n \\ &= 1 - [q^{[y]+1}]^n \\ P[Y = y] &= P[Y \leqslant y] - P[Y \leqslant y - 1]. \\ &= 1 - (q^{[y]+1})^n - 1 + (q^{[y-1]+1})^n \\ &= (q^{[y-1]+1})^n - (q^{[y]+1})^n \\ &= (q^y)^n - \left(q^{y+1}\right)^n \\ &= (q^y)^n (1 - q^n) = (q^n)^y (1 - q^n) = Q^y(1 - Q). \\ &= Q^y P \text{ where } P = (1 - q^n) \end{aligned}$$

$\text{Min}(X_i)$ also follows Geometric distribution with parameter $P = (1 - q)$.

$$Y = \min(X_i) \sim \text{Geo}(P = (1 - q^n))$$

6 Lack of Memory Property:

If X has a Geometric distribution, then $P(X > m + n \mid X > m) = P(X \geqslant n)$ $\forall$ natural no. m, n.

Proof: $p(X = x) = \begin{cases} pq^x, & x = 0, 1, 2, \ldots, \quad 0 < p < 1, p + q = 1 \\ 0 & \text{otherwise} \end{cases}$

$$\begin{aligned} p(X > k) &= \sum_{x=k+1}^{\infty} pq^x = p\left[q^{k+1} + q^{k+2} + \cdots\right] \\ &= p\frac{q^{k+1}}{1-q} = q^{k+1} \end{aligned} \tag{i}$$

Then,

$$\begin{aligned} P[X > m + n \mid X > m] &= \frac{P(X > m + n, X > m)}{P(X > m)} \\ &= \frac{P(X > m + n)}{P(X > m)} \\ &= \frac{q^{m+n+1}}{q^{m+1}} \\ &= q^n = q^{n-1+1} = P(X > n - 1) = P(X \geqslant n) \end{aligned}$$

Interpretation

This theorem states that the probability that more than $(m+n+1)$ trails required for the 1st success given that there have been more than m failures is equal to the unconditional probability of at least n trails are needed before the first success ie. $P(X > m + n \mid x > m)$ is independent of m i.e. the information of no of success in (m+1) trails has been forgotten in the subsequent calculation. This property of Geometric distribution is known as **Lack of Memory Property**.

7 Theorem

Let X be a non-negative integer valued random variable satisfying $P(X > m+n \mid X > m) = P(X \geqslant n) \forall m, n \in \mathbb{N}$. Then X must have a Geometric Distribution.

Let $p_k = p(X = k), k = 0, 1, 2, \ldots$ be the PMF of X.
Define $q_m = p(X > m) = \sum_{k=m+1}^{\infty} p_k$
Then,

$$\begin{aligned} & p(X > m+n \mid X > m) = p(X \geqslant n) \\ \Rightarrow \quad & \frac{p(X > m+n)}{p(X > m)} = p(X \geqslant n) \\ \Rightarrow \quad & p(X > m+n) = p(X \geqslant n) p(X > m) \\ \Rightarrow \quad & q_{m+n} = q_{n-1} q_m \end{aligned}$$

Set $n = 1, q_{m+1} = q_m q_0 \quad \Rightarrow \frac{q_{m+1}}{q_m} = q_0$
Now,

$$\begin{aligned} q_m &= \frac{q_m}{q_{m-1}} \frac{q_{m-1}}{q_{m-2}} \frac{q_{m-2}}{q_{m-3}} \cdots \frac{q_1}{q_0} q_0 \\ &= q_0 q_0 \ldots q_0 q_0 \\ &= q_0^{m+1} \end{aligned}$$

Again,

$$\begin{aligned} q_{m-1} &= p_m + p_{m+1} + p_{m+2} + \ldots \\ -q_m &= -p_{m+1} - p_{m+2} - p_{m+3} + \cdots \\ \hline q_{m-1} - q_m &= p_m \end{aligned}$$

$\therefore p_m = q_{m-1} - q_m = q_0^m - q_0^{m+1} = q_0^m (1 - q_0)$
Now, $q_0 = P(X > 0) = 1 - P(X \leqslant 0) = 1 - p_0$
So, $p_m = \begin{cases} p_0 (1-p_0)^m, & m = 0, 1, 2, \ldots \\ 0, & \text{otherwise} \end{cases}$
$\therefore X \sim \text{Geo } (p_0)$.

Remark

Among all non-negative integer valued R.V's X has the lack of memory property iff X has a Geometric distribution. Hence the lack of memory property is a characterization of Geometric Random variable in the class of all nonnegative integer valued Random variable.

2. Show that $E\left(\frac{1}{1+X}\right) = \sum_{x=0}^{\infty} \frac{1}{1+x} p q^x = -\frac{p}{q} \log p$.

$$\begin{aligned} E\left(\frac{1}{1+X}\right) &= \sum_{x=0}^{\infty} \frac{1}{1+x} p q^x \\ &= \frac{p}{q} \sum_{x=0}^{\infty} \frac{1}{1+x} q^{x+1} \\ &= \frac{p}{q} \left\{ \frac{q^1}{1} + \frac{q^2}{2} + \frac{q^3}{3} + \cdots \right\} \\ &= -\frac{p}{q} \left\{ -q - \frac{q^2}{2} - \frac{q^3}{3} \cdots\cdots \right\} \\ &= -\frac{p}{q} \log(1-q) = -\frac{p}{q} \log p. \end{aligned}$$